

Question 1

Question ID: 1587926

Which of the following statements is *most* accurate about a 3-for1 stock split in one of the securities in a price-weighted index?

A) The relative weights of the other securities in the index will increase.



B) The index value will decrease because the split security's price will decrease.



C) The index divisor will have to be increased to offset the price decrease of the split security.



Explanation

A split in one security will reduce its price and therefore its weight in the index. As a result, all other securities whose prices have not changed will increase their relative weight in the index. The index value will need to be *maintained* at the pre-split level. To maintain the index value, the index divisor will have to be adjusted *down* because the numerator is also adjusted down.

(Module 42.1, LOS 42.e)

Question 2

Question ID: 1587935

Can a firm that operates in a price-taker industry earn returns greater than its cost of capital in the long run?

A) No.



B) Yes, if the firm is sustainably a low-cost producer.



C) Yes, if the firm differentiates its product from its competitors' products.



Explanation

If a producer can sustain a cost advantage over time relative to its competitors, it can earn economic profits (returns greater than the cost of capital) even in a price-taker market. In such a market, each producer typically sells a product that is nearly identical to those of its competitors (i.e., little to no product differentiation).

(Module 12.2, LOS 12.b) and (Module 45.2, LOS 45.c)

Question 3

Question ID: 1587936

An analyst states, "Smith Company has a significantly higher degree of operating leverage than its industry peers." This statement is *most likely* based on analyzing Smith's costs by their:

A) nature.



B) function.



C) relationship with output.



Explanation

Operating leverage results from having fixed operating costs. Classifying costs as fixed or variable refers to their relationship with output.

(Module 45.2, LOS 45.d)

Question 4

Question ID: 1587938

A country's banking industry consists of five banks, each with a 20% market share. One of the banks acquires a significant component of one of the other banks' operations, increasing its market share to 30% and decreasing the other bank's market share to 10%. As a result of this transaction, the Herfindahl-Hirschman Index (HHI) for the country's banking industry will:

A) increase.



B) decrease.



C) remain unchanged.



Explanation

The HHI will increase. Initial HHI = $20^2 + 20^2 + 20^2 + 20^2 + 20^2 = 2,000$.

Post-transaction HHI = $30^2 + 20^2 + 20^2 + 20^2 + 10^2 = 2,200$.

(Module 12.3, LOS 12.e) and (Module 46.1, LOS 46.c)

Question 5

Question ID: 1587939

An analyst engaging in PESTLE analysis of an industry is *most likely* to evaluate its:

A) liquidity pressures.



B) technological influences.



C) equity market capitalization.



Explanation

PESTLE analysis refers to evaluating the outside influences on an industry: political, economic, social, technological, legal, and environmental.

(Module 46.2, LOS 46.d)

Question 6

Question ID: 1587940

A historical base rate convergence approach is *most likely* to be appropriate for forecasting the sales growth of a(n):

A) online travel agency.



B) chain of grocery stores.



C) high-fashion clothing company.



Explanation

The historical base rate convergence approach is best suited for companies in non-cyclical or well-established industries. A grocery chain likely faces demand for its products that does not fluctuate significantly with economic cycles. An online travel agency or a high-fashion clothing company most likely faces more cyclical demand for their products.

(Module 47.1, LOS 47.b)

Question 7

Question ID: 1587941

Grovers Inc. manufactures children's toys in a country where it has a significant share of the market and barriers to entry are high. Proxe Corp. manufactures underwear globally in a highly competitive environment. Recent innovations in machinery have reduced operating expenses in both Grovers's and Proxe's industries. The reduced operating expenses are *most likely* to:

A) increase operating margins for both companies.



B) increase operating margins more for Grovers than for Proxe.



C) have little impact on the operating margins of either company.



Explanation

Because Grovers operates in a market with weaker competitive forces than Proxe, Grovers likely has more pricing power than Proxe. **Therefore**, Grovers can likely keep its pricing unchanged so that its lower operating costs will increase its operating margins. By contrast, in Proxe's market, competitors are more likely to respond to decreased operating expenses by decreasing their prices and attempting to gain market share. Proxe will likely need to decrease its prices to defend its market share, which will keep pressure on its operating margins.

(Module 46.2, LOS 46.d) and (Module 47.1, LOS 47.c)

Question 8

Question ID: 1587946

AgroCorp Inc., a Canadian grain producer, recently issued a five-year floating rate note with a face value of CAD 90,000,000. The bond pays semiannual interest based on the Canadian Overnight Repo Rate (CORRA) plus 150 bps per year. If CORRA is 0.50% in the final interest period, at maturity AgroCorp will pay bondholders:

A) CAD 1,800,000.



B) CAD 90,900,000.



C) CAD 91,800,000.



Explanation

At maturity, AgroCorp will have to pay both the final coupon and the principal value of the bond to investors. The coupon rate is $\text{CORRA} + 0.50\% = 2.00\%$ annually. For the last six-month period, the coupon interest is $2\% / 2 \times \text{CAD } 90,000,000 = \text{CAD } 900,000$. Therefore, AgroCorp will have to pay bondholders $\text{CAD } 90,000,000 + \text{CAD } 900,000 = \text{CAD } 90,900,000$.

(Module 49.1, LOS 49.a)

Question 9

Question ID: 1587961

Following a one-notch downgrade by Standard & Poor's of a corporate bond from investment grade to non-investment grade, the bond would carry a credit rating of:

A) BBB–, and the bond yield would likely increase.



B) BB+, and the bond yield would likely increase.



C) BB+, and the bond yield would likely decrease.



Explanation

A one-notch downgrade from investment grade to non-investment grade (high-yield) rating implies a downgrade from BBB– to BB+. A rating downgrade typically increases the bond's yield.

(Module 51.1, LOS 51.a) and (Module 62.1, LOS 62.b)

Question 10

Question ID: 1587948

BAMS Brokers is looking to enter a reverse repurchase agreement involving a security that is difficult to borrow. In this scenario, BAMS would:

A) buy the security, and the repo rate is likely to be below the general collateral repo rate.



B) sell the security, and the repo rate is likely to be above the general collateral repo rate.



C) buy the security, and the repo rate is likely to be above the general collateral repo rate.



Explanation

To carry out a reverse repo, BAMS is the buyer of the security (collateral) and therefore lends money; from the seller's side, this is a repo agreement and the seller is the borrower. Because the security is difficult to borrow (special collateral), the repo rate is likely to be below the general collateral repo rate.

(Module 52.1, LOS 52.b)

Question 11

Question ID: 1587953

Lando Bank recently issued a 3-year floating rate note with quarterly interest of $\text{MRR} + 0.60\%$. The MRR is currently 0.40% . Six months later, the required margin is 40 bps (0.40%). On reset date, the floating rate note will *most likely* be priced at:

A) par.



B) a discount.



C) a premium.



Explanation

The required margin declined from 60 bps to 40 bps, indicating that Lando's credit risk has improved (the market now requires a lower spread above the MRR). Because the required margin (40 bps) is below the quoted margin (60 bps), the floating rate note will be trading at a premium.

(Module 56.1, LOS 56.a)

Question 12

Question ID: 1587957

A fixed income trader expects that the yield of a 5-year, 5% annual coupon bond with a Macaulay duration of 4.25 will decrease from 7% to 6%. As a result, the anticipated percentage increase in the bond's price is:

A) 4.25%.



B) less than 4.25%.



C) more than 4.25%.



Explanation

We can estimate the bond's price change based on modified duration. We can calculate modified duration by dividing Macaulay duration by the yield:

$$\text{ModDur} = \frac{\text{MacDur}}{(1+r)} = \frac{4.25}{1.06} = 4.009$$

As a result, the estimated price increase of the bond for a 100 bps decrease in yield is 4.009%.

(Module 59.1, LOS 59.a)

Question 13

Question ID: 1587958

Consider the following bonds, all option-free and from the same issuer:

- Bond X: 4-year, zero coupon bond
- Bond Y: 10-year, zero coupon bond
- Bond Z: 10-year, 5% coupon bond

Which of the bonds is *most likely* to have the greatest convexity?

A) Bond X.



B) Bond Y.



C) Bond Z.



Explanation

Similar to duration, a bond's convexity increases as its time-to-maturity increases and the coupon rate decreases. As a result, a long-term, zero-coupon bond such as Bond Y would have greater convexity than a coupon bond of the same maturity or a shorter-term zero-coupon bond.

(Module 60.1, LOS 60.a)

Question 14

Question ID: 1587965

A pension fund manager is analyzing the long-term bonds of three countries. The manager wants to invest in the bonds of the country with the strongest credit profile.

Characteristics	Country A	Country B	Country C
Real GDP growth (%)	0.50	0.75	1.00
Long-term external debt/GDP (%)	53.2	62.5	48.1
Short-term debt / Long-term debt (%)	35.0	24.7	28.9
FX reserves/GDP (%)	17.0	20.5	20.1

Based on the table above, the manager would *most likely* select:

A) Country A.



B) Country B.



C) Country C.



Explanation

Country C would most likely have the highest credit quality because it has the lowest long-term external debt-to-GDP ratio. Although Country B has marginally higher FX reserves-to-GDP ratio and lower refinancing risk (lower relative short-term debt), it has a significantly higher external debt burden and slightly weaker economic growth.

Question 15

Question ID: 1587966

Azzio Corp. is a frequent debt issuer and issues debt with various priority of claims. An investor who wants to invest in an Azzio bond that has the highest priority of claims would *most likely* purchase its:

A) 10-year, second lien debt.



B) 6-month, senior unsecured debt.



C) 20-year, senior unsecured debt.



Explanation

Second lien debt is a secured debt obligation, and as such ranks ahead of all unsecured obligations. All senior unsecured creditors would be treated as the same class in bankruptcy, regardless of debt maturity. Therefore, bondholders of the 6-month and 20-year senior unsecured debt have the same priority of claims.

(Module 64.1, LOS 64.c)

Question 16

Question ID: 1587967

TKA Industries (TKAI) is a European semiconductor manufacturer with €2.5 billion senior unsecured corporate bonds outstanding. In 20X1, TKAI sold €1 billion of its balance sheet receivables to TKA Trust, a special purpose entity, which then issued €800 million senior bonds and €200 million subordinated bonds. In 20X3, TKAI issued a further €1.3 billion senior unsecured corporate bonds. Which of the following statements about these bonds' seniority is *most* accurate? TKAI's:

A) €1.3 billion corporate bonds are senior to TKA Trust's subordinated bonds.



B) €2.5 billion corporate bonds have the same seniority as TKA Trust's senior bonds.



C) €1.3 billion corporate bonds have the same seniority as its €2.5 billion corporate bonds.



Explanation

TKAI's €2.5 billion and €1.3 billion senior unsecured obligations have the same priority of claims and therefore seniority, regardless of when they were issued and maturity. Seniority does not apply between TKA Trust and TKAI because they are two distinct entities; TKA Trust is an SPE and is therefore bankruptcy remote from TKAI.

(Module 64.1, LOS 64.c) and (Module 65.1, LOS 65.b)

Question 17

Question ID: 1587968

Consider the following 5-year asset-backed notes issued by Zollo Trust:

Tranche	Face value	Interest rate	Credit enhancement
Class A notes	\$80 million	MRR + 0.2%	-
Class B notes	\$10 million	MRR + 1.0%	Subordinated to Class A notes
Class C notes	\$7 million	MRR + 2.0%	Subordinated to Class A and Class B notes
Class D notes	\$3 million	MRR + 3.0%	Subordinated to Class A, Class B and Class C notes

During the year 20X3, MRR was 2.0% and defaults in the pool were \$15 million. An investor who held \$5 million of Class A notes and \$5 million of Class B notes would receive interest for the year *closest* to:

A) \$110,000.



B) \$185,000.



C) \$260,000.



Explanation

Given a default of \$15 million in the pool, Class D notes would absorb the first \$3 million in losses, Class C notes would absorb the next \$7 million, and Class B notes would absorb the remaining \$5 million in losses. Because half the pool supporting the Class B notes is in default, each investor in the Class B notes receives half of the scheduled interest payments. Investors in Class A notes receive full interest payments.

Interest received for Class A notes: $\$5 \text{ million} \times (2.0\% + 0.2\%) = \$110,000$

Interest received for Class B notes: $1/2 \times \$5 \text{ million} \times (2.0\% + 1.0\%) = \$75,000$

Total interest received = $\$110,000 + \$75,000 = \$185,000$

(Module 66.1, LOS 66.b)

Question 18

Question ID: 1587969

A sequential-pay collateralized mortgage obligation (CMO) has three tranches: Tranche A matures in 2 years, Tranche B matures in 5 years, and Tranche C matures in 10 years. An investor who expects a gradual decline in interest rates over the next 5 years would *most likely* be concerned about:

A) contraction risk, and therefore would prefer Tranche A.



B) extension risk, and therefore would prefer Tranche B.



C) contraction risk, and therefore would prefer Tranche C.



Explanation

The investor would be concerned about contraction risk, or the risk that decreasing interest rates will cause interest payments to decline and principal to be prepaid more quickly than anticipated over the next 5 years. The strongest protection against contraction risk in a sequential-pay CMO comes from the longest maturity, in this case Tranche C.

(Module 67.1, LOS 67.a, LOS 67.c)

Question 19

Question ID: 1587970

BamBam Inc. is a Canadian manufacturer of specialized machinery to produce pet food. BamBam exports its machinery to clients in many different countries. A recent transaction involves the sale of a machine to a client in France. The contract date is May 31st, the delivery date is 60 days later, and the client will make payment to BamBam in Euros. Which of the following reasons *best* explains why BamBam should use a derivative rather than a spot market transaction to manage the currency risk of the contract?

A) BamBam is selling a machine at a contract price in euros but incurs costs in Canadian dollars.



B) BamBam is unable to sell euros today to offset the contract price of the machinery delivered to the French client.



C) BamBam faces a 60-day timing difference between the contract date and the delivery date when it is paid for the machine by the French client.



Explanation

A 60-day timing difference exists between the contract and the delivery date when BamBam is paid for the machine in EUR. A derivative contract in which the underlying EUR/CAD forward rate is agreed upon today and exchanged on the delivery date allows BamBam to hedge or offset the CAD value of the future EUR payment. Therefore, the derivative is a more suitable contract to address the currency risk of the transaction than a spot market sale of EUR.

(Module 68.1, LOS 68.a)

Question 20

Question ID: 1587971

An investor has written a 3-month call option on an asset at \$35 for \$4 when the asset is trading at \$38. If, at expiration, the asset price is \$37, the investor will have a:

A) payoff of \$2 at expiration.



B) profit of \$2 at expiration.



C) loss of \$2 at expiration.



Explanation

A writer of the call will have a negative payoff of \$2 at expiration for a profit of $4 - 2 = \$2$. (Module 69.2, LOS 69.b)

Question 21

Question ID: 1587972

Ruby Inc. is a company in based in the United States that has recently acquired a subsidiary located in Switzerland. The company is concerned about the impact of fluctuations in the USD/CHF exchange rate on the value of the Swiss subsidiary. Which type of hedge would be *most* appropriate for Ruby to consider?

A) Cash flow hedge.



B) Fair value hedge.



C) Net investment hedge.



Explanation

A net investment hedge is designated as offsetting the foreign exchange risk of the equity of a foreign operation and can be done with a currency swap or forward, for example. A cash flow hedge is meant to absorb the variable cash flow of a floating-rate asset or liability. A fair value hedge is meant to offset fluctuations in the fair value of an asset or liability.

(Module 70.1, LOS 70.b)

Question 22

Question ID: 1587973

Which of the following *least likely* illustrates the correct way to calculate the forward price at time T of an asset (S) that has both costs and benefits of holding the asset, in addition to the opportunity cost of funds (Rf)?

A) $[S_0 + PV_0(\text{benefits}) - PV_0(\text{costs})] (1 + Rf)^T$



B) $[S_0 - PV_0(\text{benefits}) + PV_0(\text{costs})] (1 + Rf)^T$



C) $S_0(1 + Rf)^T + FV^T(\text{costs}) - FV^T(\text{benefits})$



Explanation

The benefits of holding the asset are subtracted and the costs of holding the asset are added when calculating the no-arbitrage forward price.

$[S_0 - PV_0(\text{benefits}) + PV_0(\text{costs})](1 + Rf)^T = S_0(1 + Rf)^T + FV^T(\text{costs}) - FV^T(\text{benefits})$, and either can be used to calculate the no-arbitrage forward price.

(Module 71.1, LOS 71.b)

Question 23

Question ID: 1587974

Assuming all other parameters remain unchanged, what is the impact of a decrease in the risk-free rate after the inception of a forward contract on the contract's mark-to-market value to the forward buyer? The forward contract's mark-to-market value to the forward buyer will:

A) increase.



B) decrease.



C) remain unchanged.



Explanation

The mark-to-market value from the forward buyer's perspective is equal to $V_t(T)$ as follows:

$$V_t(T) = S_t - F_0(T)(1 + r)^{-(T-t)}$$

A decrease in the risk-free rate following the inception of a forward contract will increase the present value of the forward price $F_0(T)$, which will decrease the mark-to-market value from the contract buyer's perspective.

(Module 72.1, LOS 72.a)

Question 24

Question ID: 1587975

The price of a forward contract:

A) is stated in the contract.



B) equals zero at contract initiation.



C) may fluctuate during the contract's life.



Explanation

The *price* of a forward contract is the price at which the underlying asset will be exchanged at settlement. This price is stated in the contract and determined such that the *value* of the contract is equal to zero at initiation. The value may vary during the life of the contract, but the price does not.

(Module 73.1, LOS 73.a)

Question 25

Question ID: 1587976

At initiation of a forward contract and a futures contract with identical terms, their prices are *most likely* to be different if:

A) the spot price is highly volatile.



B) the forward contract is marked to market daily.



C) short-term interest rates are negatively correlated with futures prices.



Explanation

Forward and futures prices may differ for otherwise identical contracts if interest rates are positively or negatively correlated with futures prices. The difference arises from the fact that futures are marked to market daily while forwards typically are not. A futures contract holder can earn interest on mark-to-market gains and faces an opportunity cost of interest of mark-to-market losses.

(Module 73.1, LOS 73.b)

Question 26

Question ID: 1587977

At inception, a one-year quarterly pay fixed-for-floating interest rate swap:

A) has four unknown future payments.



B) has expected payments with a present value of zero.



C) can be replicated with a series of forward rate agreements at market rates.



Explanation

At the inception of a swap, the present value of the expected cash flows from the swap is zero. The fixed and floating rates for the first payment are known at inception, so the net payment at the end of the first period is known at inception. We can create a position equivalent to the swap with market-rate FRAs, but it will not replicate the swap because the fixed rates on the market-rate FRAs are not all equal.

(Module 74.1, LOS 74.a)

Question 27

Question ID: 1587978

The price of an interest rate swap is *most accurately* described as:

A) zero at the initiation of the swap contract.



B) decreasing when the reference rate increases.



C) constant throughout the life of the swap contract.



Explanation

The price of an interest rate swap is the fixed interest rate specified in the contract. The value of an interest rate swap is typically zero at initiation and may vary during the life of the contract as expected future interest rates change.

(Module 74.1, LOS 74.b)

Question 28

Question ID: 1587979

Other things equal, cash dividends on a stock during an option's life:

A) decrease the value of call options and put options on the stock.



B) increase the value of call options on the stock and decrease the value of put options on the



C) stock.



D) decrease the value of call options on the stock and increase the value of put options on the stock.



Explanation

Dividend payments decrease the price of the underlying stock. This makes calls less valuable and puts more valuable.

(Module 75.1, LOS 75.c)

Question 29

Question ID: 1587980

European call and put options with an exercise price of 60 and an expiration date of August 31 are available on an underlying asset. The present value of the price of a forward contract on the same asset today is *closest* to the:

A) call price minus the put price plus the present value of 60.



B) put price minus the call price plus the present value of 60.



C) call price plus the put price minus the present value of 60.



Explanation

The put-call-forward parity relationship states that at the initiation of a forward contract, $F_0(T) (1 + R_f)^{-T} = c_0 - p_0 + X (1 + R_f)^{-T}$. In this question, the exercise price, X , is 60.

(Module 76.1, LOS 76.b)

Question 30

Question ID: 1587981

In a one-period binomial model of an option that uses probabilities of an up-move and a down-move, the probabilities are:

A) assumed to be equal to 0.5.



B) estimated based on historical data.



C) calculated from the model's inputs.



Explanation

The risk-neutral probabilities used in a one-period binomial model for options are derived from the risk-free rate and the sizes of the up-moves and down-moves in the model.

(Module 77.1, LOS 77.b)

Question 31

Question ID: 1587937

Nagle Company produces a commodity with a market price of \$3.25 per unit. Nagle forecasts production and sales of 900,000 units over the next quarter. Nagle's variable cost per unit is \$3.75, and fixed operating costs are \$400,000. Based on these projections, it is *most likely* that:

A) Nagle's contribution margin for the quarter is \$0.50 per unit.



B) Nagle will experience an operating loss of \$400,000 for the quarter.



C) Nagle's breakeven quantity of sales for the quarter is 1,200,000 units.



Explanation

Nagle has no breakeven quantity of sales because the price per unit is less than Nagle's variable cost per unit (i.e., the contribution margin is negative). Nagle's operating income would be $900,000(\$3.25 - \$3.75) - \$400,000 = -\$850,000$ by operating in the next period, but Nagle can limit its operating loss to its fixed operating costs of \$400,000 by shutting down temporarily.

(Module 12.1, LOS 12.a) and (Module 45.2, LOS 45.d)

Question 32

Question ID: 1587923

A dark pool is *best* described as:

A) an exchange with low trading volume.



B) the loans underlying a mortgage-backed security.



C) an alternative trading system that does not reveal current orders.



Explanation

An alternative trading system that does not reveal current client orders is referred to as a dark pool. Dark pools provide traders anonymity by not revealing their orders to potential counterparties.

(Module 41.1, LOS 41.d)

Question 33

Question ID: 1587924

Jacques Fontenot wants to place an order to purchase 10,000 shares of BQ Inc. at a price of €75.00 or below. The shares are currently trading for €82.10 bid and €82.20 ask. What type of order should Fontenot place?

A) Market order.



B) Stop loss order.



C) Limit order.



Explanation

This is a limit order. If the shares trade for €75 or below, Fontenot's broker will purchase the shares for him. If Fontenot had wanted to sell shares he already owned when the price dropped to €75.00, that would be a stop loss order.

(Module 41.3, LOS 41.h)

Question 34

Question ID: 1587925

In a call market:

A) a single price that clears the market is set periodically.



B) trades may occur at any time during market hours.



C) prices are set by the highest dealer bid and the lowest dealer ask price.



Explanation

Trades in a call market occur at specified times. All orders are accumulated, and a single negotiated price is set that clears the market.

(Module 41.3, LOS 41.j)

Question 35

Question ID: 1587927

A mutual fund uses a momentum strategy-buying stocks that have been increasing in price and selling stocks that have been decreasing in price. The *most appropriate* type of equity index to use as a benchmark is:

A) equal weighted.



B) fundamental weighted.



C) market capitalization weighted.



Explanation

In a market capitalization weighted index, the relative weight increases for constituent stocks that have outperformed other stocks in the index. For this reason, a market capitalization weighted index is an appropriate benchmark for a fund using a momentum strategy (buying securities that have been increasing in price and selling securities that have been decreasing in price). Fundamental weighted indices often have a value tilt, not a momentum tilt. With equal weighting there is no tilt toward momentum.

(Module 42.1, LOS 42.d)

Question 36

Question ID: 1587928

Sam Grant, CFA, publishes the Grant 50 Index, which consists of value stocks with small market capitalizations. The Grant 50 Index is best described as a:

A) style index.



B) sector index.



C) value weighted index.



Explanation

Style indexes measure the returns to strategies that are differentiated by market capitalization or by value versus growth. A sector index measures the returns for an industry sector such as financials. Although market capitalization is one of the criteria for inclusion in the index, this does not suggest the index is value weighted.

(Module 42.2, LOS 42.j)

Question 37

Question ID: 1587929

A market researcher is analyzing the efficiency of the Oceania Securities Exchange (OSE). Market prices appear to incorporate all prior price and volume information in a timely manner but are slow to incorporate the true value impact of earnings surprises. The researcher determines that specialists on the floor of the OSE consistently earn positive risk-adjusted returns on average. Which of the three forms of the Efficient Market Hypothesis *best* describes the OSE market?

A) Weak-form efficient.



B) Strong-form efficient.



C) Semistrong-form efficient.



Explanation

In a weak-form efficient market, market prices incorporate all market related information. In a semistrong-form efficient market, market prices incorporate all market and non-market public information. In a strong-form efficient market, market prices incorporate all market, non-market public, and private information. Positive risk-adjusted returns for specialists indicate that this market is not strong-form efficient as specialists have access to non-public information. Slow price adjustment to earnings surprises suggest this market is not semistrong-form efficient.

(Module 43.1, LOS 43.d)

Question 38

Question ID: 1587930

Participating preference shares are *most accurately* described as preferred stock that:

A) may be exchanged for the firm's common shares at a predetermined ratio.



B) receives extra dividends if the firm's profits are greater than a predetermined level.



C) must be paid any omitted dividends from prior periods before the firm may pay dividends to common shareholders.



Explanation

Participating preference shares pay extra dividends if the firm's profits exceed a predetermined level. Convertible preference shares may be exchanged for the firm's common shares at a predetermined ratio. Cumulative preferred shares must be paid any omitted dividends from prior periods before the firm may pay dividends to common shareholders.

(Module 44.1, LOS 44.a)

Question 39

Question ID: 1587931

Compared with publicly traded firms, private firms *most likely* tend to:

A) focus more on long-term performance.



B) adopt stronger corporate governance policies.



C) be more closely regulated and have higher compliance costs.



Explanation

Private equity firms tend to be better able to focus on long-term performance than publicly traded firms because they do not face pressure from public shareholders to produce positive short-term results. Because their shares are not publicly traded, private equity firms are not required to comply with the same financial reporting regulations as publicly traded firms. As a result, private equity firms have lower compliance costs than comparable publicly traded firms, but less scrutiny from public shareholders may result in weaker corporate governance.

(Module 44.1, LOS 44.c)

Question 40

Question ID: 1587932

Global depository receipts (GDRs) are:

A) predominantly priced in U.S. dollars.



B) traded in United States markets.



C) sponsored by the company issuing shares.



Explanation

Global depository receipts (GDRs) are typically denominated in U.S. dollars. GDRs are not listed on United States securities exchanges. Depository receipts that trade in the United States are called American depository receipts (ADRs). GDRs may be sponsored by the issuing firm or unsponsored and issued by the depository bank.

(Module 44.2, LOS 44.d)

Question 41

Question ID: 1587933

Other things equal, preferred stock is *most likely* to pay the highest dividend yield if it has which of the following features?

A) Puttable.



B) Callable.



C) Option-free.



Explanation

Because callable preferred stock has limited upside price potential, investors typically require it to pay a higher dividend yield than otherwise equivalent option-free or puttable preferred stock.

(Module 44.2, LOS 44.e)

Question 42

Question ID: 1587912

The present value of an equity security's future cash flows is *most likely* to be significantly different from the company's:

A) book value per share.



B) market value per share.



C) intrinsic value per share.



Explanation

A company's book value per share is typically not equal to its intrinsic value per share, since book value is based on the historical cost of assets. Intrinsic value is the present value of the security's future cash flows. The market value of a share is the cumulative investors' estimation of this intrinsic value, and market value is usually somewhat in line with the intrinsic value.

(Module 44.2, LOS 44.g)

Question 43

Question ID: 1587934

Which of the following is *most likely* to increase a firm's return on equity (ROE)?

A) Issuing new equity to retire debt.



B) Issuing new debt to retire common stock.



C) A decrease in the market value of its common stock.



Explanation

Because the after-tax cost of debt is typically less than the earnings yield on the stock, issuing debt and repurchasing common stock is most likely to increase ROE, although the variability of net income will be increased (increased financial leverage). Changes in the market value of the common stock will not affect ROE.

(Module 44.2, LOS 44.h)

Question 44

Question ID: 1587942

An analyst uses a temporary supernormal growth model to value a common stock. The company paid a \$2 dividend last year. The analyst expects dividends to grow at 15% each year for the next three years and then to resume a normal rate of 7% per year indefinitely. The analyst estimates that investors require a 12% return on the stock. The value of this common stock is *closest* to:

A) \$39.



B) \$53.



C) \$65.



Explanation

Calculate the dividends during the supernormal growth period using $g_s = 15\%$.

$$D_1 = D_0(1 + g_s) = \$2.00(1.15) = \$2.30$$

$$D_2 = D_0(1 + g_s)^2 = \$2.00(1.15)^2 = \$2.645$$

$$D_3 = D_0(1 + g_s)^3 = \$2.00(1.15)^3 = \$3.042$$

D_3 is the first dividend that *will grow* at a constant rate. Use this dividend to calculate the value of the stock at $t = 2$ using the infinite period DDM.

$$P_2 = \frac{D_3}{(k_e - g)} = \frac{3.042}{0.12 - 0.07} = 60.84$$

Calculate the present value of the cash flows discounted at k_e of 12%.

$$\text{PV of } D_1 = \$2.30 / (1.12) = \$2.054$$

$$\text{PV of } D_2 = \$2.645 / (1.12)^2 = \$2.109$$

$$\text{PV of } P_2 = \$60.84 / (1.12)^2 = \underline{\$48.50}$$

$$V_s = \$52.664$$

(Module 48.2, LOS 48.h)

Question 45

Question ID: 1587943

In valuing the stock of Evergreen Enterprises, an analyst compiles the following information about the firm:

Expected constant growth rate of dividends	6%
Next year's expected earnings per share	\$4.24
Expected retention ratio	62.5%
Required rate of return	11%

The value of the firm's stock today is *closest* to:

A) \$31.80.



B) \$38.55.



C) \$53.00.



Explanation

The dividend payout ratio is $1 - \text{expected retention ratio} = 1 - 0.625 = 0.375$ or 37.5%.

The leading (forward) P/E ratio is:

$$\frac{P_0}{E_1} = \frac{D_1/E_1}{r-g} = \frac{0.375}{0.11-0.06} = 7.50$$

Multiplying the leading P/E ratio times forecast earnings per share gives the stock value:

$$P_0 = (P_0 / E_1)(E_1) = 7.50(\$4.24) = \$31.80.$$

(Module 48.3, LOS 48.k)

Question 46

Question ID: 1587944

Enterprise value is *most appropriately* interpreted as:

A) the cost of a firm to a potential acquirer.



B) the estimated fair value of a firm's tangible and intangible assets.



C) the total market value of a firm's outstanding debt and equity securities.



Explanation

Enterprise value (market value of a firm's equity and debt less the firm's cash and short-term securities investments) can be interpreted as the amount that would be needed to acquire a firm. Asset-based models are used to estimate the fair value of a firm's assets.

(Module 48.3, LOS 48.I)

Question 47

Question ID: 1587945

A drawback of using the price to book value ratio as a valuation tool is that book value:

A) is not appropriate for valuing firms with primarily financial assets.



B) may not be an accurate indicator of the value of a company's assets and equity.



C) is ineffective in valuing companies that are not expected to continue as going concerns.



Explanation

An important drawback of the price to book value ratio is that book values do not necessarily reflect market values. Book values of assets reported at amortized cost can be too low if prices have increased since the company acquired the assets. Book values can also be too high, for example, if technological changes have made some of a company's assets obsolete. The price to book value ratio is useful for valuing companies that primarily hold financial assets and for valuing companies that are expected to cease operations.

(Module 48.3, LOS 48.f)

Question 48

Question ID: 1587947

Cash flows from a fully amortizing security include:

A) no balloon payment.



B) a single balloon payment.



C) a series of equal balloon payments.



Explanation

A balloon payment is a lump sum return of principal at maturity. Fully amortizing securities are structured with equal periodic payments that include both interest and principal, so that there is no balloon payment at maturity.

(Module 50.1, LOS 50.a)

Question 49

Question ID: 1587950

The difference between on-the-run and off-the-run U.S. Treasury securities is that on-the-run Treasury securities are:

A) traded only in the primary market, while off-the-run Treasury securities are traded only in the secondary market.



B) generally less actively traded than off-the-run Treasury securities and provide less reliable market yields.



C) the most recently auctioned Treasury securities in each maturity, while off-the-run Treasury securities are issues auctioned previously.



Explanation

On-the-run Treasury securities are the most recently auctioned issues. After they are auctioned in the primary market, on-the-run Treasury securities trade in the secondary market alongside off-the-run Treasury securities. On-the-run issues are generally the most actively traded Treasury securities.

(Module 53.1, LOS 53.b)

Question 50

Question ID: 1587949

For a repurchase agreement, other things equal, when the security is of higher credit quality, then

A) both the repo rate and repo margin are lower.



B) the repo rate is higher but the repo margin is lower.



C) the repo rate is lower but the repo margin is higher.



Explanation

Compared with using a security with a lower credit rating, using a security with a higher credit rating will result in a lower repo rate (smaller difference between selling price and repurchase price) and a lower repo margin [the percentage that the sale price (i.e., the amount loaned) is less than the market value of the security].

(Module 52.1, LOS 52.b)

Question 51

Question ID: 1587954

Which of the following 2-year bonds *most likely* has a yield to maturity that can be interpreted as a spot rate?

A) Price = 88.36 percent of par, yield to maturity = 6.38%.



B) Price = 100.00 percent of par, yield to maturity = 5.19%.



C) Price = 103.85 percent of par, yield to maturity = 4.72%.



Explanation

Only bonds with a single payment (that is, a pure discount bond) will have a YTM that can be interpreted as a spot rate. Of the three choices given, only the bond priced at 88.36 can be such a security at positive interest rates. The two bonds not trading at a discount must have a coupon payment at the end of the first year, so their YTMs cannot be interpreted as spot rates.

(Module 57.1, LOS 57.a)

Question 52

Question ID: 1587951

An investor bought a \$100,000 6% semiannual-pay bond for settlement on November 12, 20X3 at a full price of \$94,255. The bond makes coupon payments on January 1 and July 1 and accrued interest is calculated on a 30/360 basis. The flat price of the bond is:

A) less than \$92,060.



B) equal to \$92,060.



C) greater than \$92,060.



Explanation

Days between July 1 and November 12 = $30 + 30 + 30 + 30 + 12 = 132$. Accrued interest at settlement is $132 / 180 \times 3,000 = 2,200$. The flat price is $94,255 - 2,200 = 92,055$.

(Module 54.1, LOS 54.a)

Question 53

Question ID: 1587952

A bond has a Z-spread of 1.6% and an option-adjusted spread (OAS) of 1.4%. This bond is *most likely*:

A) puttable.



B) callable.



C) convertible.



Explanation

The value of an embedded option (to the issuer, in percentage yield terms) = z-spread – OAS. In this case, we have $1.6\% - 1.4\% = 0.2\%$. The positive value to the issuer means the bond is most likely callable because the issuer is long the call (has the right to exercise). Put and conversion features have positive value to the bondholder (because the bondholder has the right to exercise them) and negative value to the issuer.

(Module 55.1, LOS 55.b)

Question 54

Question ID: 1587955

Based on the following rates:

1-year spot rate 2.0%

2-year spot rate 2.5%

3-year spot rate 3.0%

4-year spot rate 3.5%

The 2-year forward rate two years from now is *closest* to:

A) 3.25%.



B) 3.50%.



C) 4.50%.



Explanation

$$\left(\frac{(1.035)^4}{(1.025)^2} \right)^{\frac{1}{2}} - 1 = 4.51\%$$

Alternatively, $(4 \times 3.5 - 2 \times 2.5) / 2 = 4.5\%$.

(Module 57.1, LOS 57.b)

Question 55

Question ID: 1587959

When analyzing the interest rate risk of the following bonds, which is *most likely* to require the use of effective duration?

A) 10-year, zero-coupon corporate bond.



B) 10-year, 4% coupon mortgage-backed bond.



C) 5-year, zero-coupon non-sovereign government bond.



Explanation

Effective duration is required when analyzing interest rate risk of bonds with embedded options. Because mortgage borrowers may prepay their loans, mortgage-backed bonds have a price-yield relationship similar to that of callable bonds. Therefore, analyzing their interest rate risk requires the use of effective duration. The other two bonds are option free and can be analyzed using modified duration.

(Module 61.1, LOS 61.a)

Question 56

Question ID: 1587956

When is a bond investor *most likely* to realize an annualized rate of return greater than the bond's yield to maturity (YTM) on the date of purchase? After the investor buys the bond, market interest rates:

A) increase, and the investor holds the bond to maturity.



B) decrease, and the investor holds the bond to maturity.



C) decrease, and the investor sells the bond prior to maturity.



Explanation

If market interest rates increase after the investor buys the bond, the reinvestment rate on coupon income will increase. This, in turn, will increase the annualized rate of return above the YTM at purchase if the investor holds the bond to maturity. If market interest rates decrease, this will decrease the reinvestment rate but increase the bond price. Whether this increases or decreases the annualized rate of return for an investor depends on his holding period.

(Module 58.1, LOS 58.a)

Question 57

Question ID: 1587960

Allison Coleman, CFA, owns a bond portfolio that includes Bond X, a callable bond with ten years to maturity that is callable at any time beginning one year from today. Coleman's portfolio also includes Bond Y, a noncallable security with ten years to maturity that carries the same credit rating as Bond X. Coleman expects interest rates to decrease steadily over the next few years. Based on this assumption, Coleman should expect that:

A) Bond Y will experience a larger decrease in value than Bond X.



B) Bond X will benefit from positive convexity as rates decline.



C) the option embedded in Bond X will increase in value.



Explanation

As rates fall, the option becomes more valuable, and the price of Bond X will not appreciate as rapidly as the noncallable Bond Y. The negative convexity of the callable bond will limit its price appreciation potential.

(Module 61.1, LOS 61.a)

Question 58

Question ID: 1587962

Chris South owns \$25,000 face value of Bradco bonds, which have a 7% coupon, pay interest semiannually, and have six years remaining until maturity. The bonds are callable at par. The bonds were rated A when Chris bought them at par two years ago, and they are currently worth \$26,225, with a rating of AA. Over this 2-year period, the Bradco bonds have experienced a(n):

A) decrease in call risk.



B) increase in liquidity risk.



C) decrease in credit risk.



Explanation

The risk most likely to have increased is call risk, as the bonds have appreciated well above par value, quite possibly due to falling rates, which might motivate the issuer to call the bonds and replace them with lower cost debt. Credit risk has decreased, since the bonds have improved in rating from A to AA. There is no information to suggest that liquidity risk has changed, although a higher rated bond would likely be somewhat more liquid (less liquidity risk).

(Module 62.1, LOS 62.b)

Question 59

Question ID: 1587963

The *least likely* item to be a cause for concern when using bond ratings from a commercial provider is that:

A) event risk is difficult for ratings firms to estimate.



B) lower bond ratings are less stable than higher bond ratings.



C) bond ratings are backward looking and based on prior financial results.



Explanation

Bond ratings are forward looking, to the extent possible, and attempt to assess the probability of future default. The other two statements are true.

(Module 62.1, LOS 62.b)

Question 60

Question ID: 1587964

When calculating and interpreting a firm's leverage and coverage ratios, which component of traditional credit analysis is an analyst *most likely* addressing?

A) Capacity.



B) Character.



C) Collateral.



Explanation

Leverage and coverage ratios are data for analyzing company fundamentals, which is a step in the analysis of a firm's capacity to pay its debts.

(Module 62.1, LOS 62.a)